

WAVY (Wearable hAptics for Virtual realityY)

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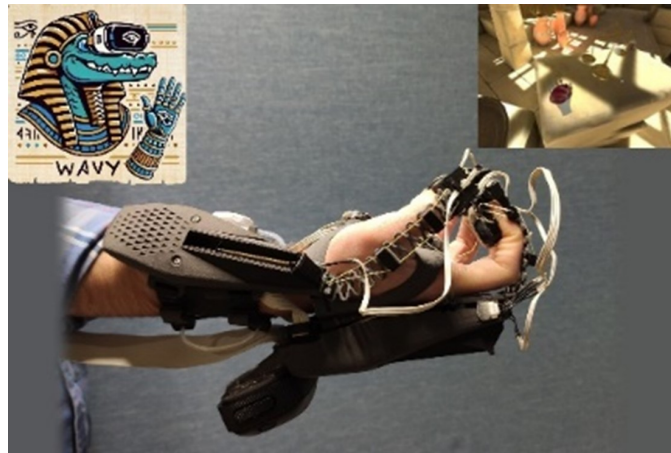


Figure 1: Haptic prototype from the WAVY project

Abstract

Despite the importance of touch in human interaction with the environment, the full potential of this sensory modality remains underutilized in human-computer interaction systems. This is mostly due to challenges related to scalability, integration, and cost. This is particularly noticeable in the field of virtual reality (VR), where the system should be portable, lightweight, highly integrated, robust and easy to set up. Haptics has become the main challenge in developing immersive experiences. For several years, researchers and industries have attempted to create haptic hand wearables but they remained too expensive for the mass market. We have designed a new actuator based on active lubrication made from low-cost components and integrated into a hand-worn device that combines tracking, multi-point vibrations and force feedback for the palm and the fingers. The device is showcased in a VR scenario, while the actuator technology can also be explored through a standalone interactive demonstration.

*All authors contributed equally to this research.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Interaction devices; Haptic devices;**

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1 Introduction

Haptics is fundamental to understand and interact with our environment. This is why haptic devices have been developed for decades in teleoperation applications [Bolopion and Régner 2013][Liu et al. 2019]. Hernandez-Gobertti et al. found that for teleoperation tasks haptic feedback enables shorter execution time mostly for complex tasks [Hernandez-Gobertti et al. 2025]. It also improves execution precision. Most of these devices are grounded, which enables to deport actuators and electronics, and in turn to provide strong force feedback and precise tracking and control. However, they can suffer from a limited workspace.

As virtual reality (VR) headset developed, the users needed to interact in a large workspace which led to wearable haptic devices



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[He, Shiyu et al. 2023]. New applications emerged, such as the teaching of technical skills [Nassar et al. 2021][Xie et al. 2021], cultural content [Theodoropoulos and Antoniou 2022] or social interaction in VR [Jacucci et al. 2024]. Subsequently, new needs emerged for wearable haptic devices such as enhancing the sense of immersion (device transparency, support for multi-modal feedback), providing affordable devices for large scale deployment and improving robustness, ease of installation and usage. Hernandez-Gobertti et al. showed that scenarios using force feedback mechanisms increased the sense of engagement of users [Hernandez-Gobertti et al. 2025]. However, new devices should not necessarily prioritize realism in the virtual world to the exclusion of all the physical and economic constraints mentioned earlier [Teng et al. 2026]. For instance, one can combine several modalities or actuators to provide haptic illusions to reduce the number of actuators. It can even be counterproductive to overload the sense of touch when trying to obtain realistic feedback.

In recent years, several portable haptic devices have been developed, but they have yet to gain a foothold in the consumer market. Consequently, haptic feedback remains a critical missing component in user interfaces compared to visual and auditory rendering that have seen major advances. This can be explained by the fact that current devices are very specific for one or two of the following four functions: force feedback (SenseGlove Nova [SenseGlove 2026], Haptx [Haptx 2026], Dexmo [DextaRobotics 2026]), vibrotactile feedback (electrotactile [Lin et al. 2022], electroosmotic actuators [15], micro-PZT stack actuators [Jin et al. 2022], servomotors [Weart 2026], pneumatic actuators [Haptx 2026]), thermal feedback (TouchDIVER Weart [18]) and finger tracking (Manus Prime II [Manus 2026]). The other functions are not available or with limited performance. When device makers try to bring high performance to more than one of these functions, it becomes complex, bulky, and costly. To address this issue, we have explored an innovative technology that significantly reduces the size and weight of the actuator responsible for providing the haptic feedback felt when closing the hand. This actuator based on active lubrication also delivers high-frequency haptic feedback, providing resistive and textured feedback during finger flexion. It uses low-cost piezoelectric actuators powered by a modulated ultrasonic signal to reduce friction between sliding surfaces [Pontreau et al. 2025a][Pontreau et al. 2025b]. The actuator acts as a brake by counteracting the force applied by the user and only provides a haptic signal when a movement is initiated. In addition to this actuator, our device includes six wide-bandwidth haptic actuators and one active actuator that can squeeze the hand.

The six actuators are required to provide tactile feedback when the fingers are not moving. They provide feedback to the palm of the hand and the fingertips. While haptic devices often provide vibrotactile feedback, they often use short bandwidth actuators like Linear Resonance Actuators (LRA) (Manus Prime II [10], SenseGlove Nova [SenseGlove 2026]), in order to reduce cost, size and complexity. There are touch-sensitive actuators available that offer high spatial resolution and high bandwidth [Jin et al. 2022][Liu et al. 2019][Shen et al. 2023], but they are still far from industrial production. In our device, we used wide bandwidth voice coil actuator in order to demonstrate the significance of haptic richness it can provide.

2 The Demonstration

The device being demonstrated was developed as part of the WAVY project. Rather than pursuing high-fidelity force feedback, WAVY focused on an innovative actuation mechanism inspired by tendon-based systems and leveraged advances in multi-modal cognitive illusions to deliver compelling and scalable haptic experiences. The prototype showcases the potential of active lubrication brakes for haptic feedback in VR environment. It demonstrates how wide-band and spatially distributed vibrotactile feedback can enhance VR experience. The device also generates active hand squeezing to show that only one DC motor can produce various perceptive illusions. The final goal of the project was to support content creation, hence real-time communication modules, rapid prototyping tools, and seamless Unity integration were developed, allowing non-expert users to easily add interactive haptic content to their virtual scenes.

During the demonstration users will be immersed in an Egyptian atmosphere and will feel a heartbeat sensation, contacts with objects on the palm and fingertips and interaction with water and insects. At the end of the scenario, the user will interact with several objects to compare their hardness, weight, impact response, and discover how multi-modal haptic feedback can provide such sensations.

The core of the demonstration is the active lubrication brake technology. This innovative braking mechanism is implemented on three fingers. Performance exceeds previously reported results in the literature, achieving up to a 95% reduction in friction force, with a controllable and repeatable force range between 0.24 N and 2.65 N with a vibro-tactile modulation up to 200 Hz. This actuator is further demonstrated in a standalone slider and a user interface example, showcasing other potential applications.

All developments (hardware prototypes, haptic effects, and sensory illusions) were evaluated with end users throughout the design cycle using an iterative, user-centered approach. This included artistic residencies involving all consortium partners and external testers, as well as public demonstrations at conferences and events. The final device design was further refined to be modular, compact, adaptable to different hand sizes, and visually appealing.

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References

- A. Bolepion and S. Régnier. 2013. A Review of Haptic Feedback Teleoperation Systems for Micromanipulation and Microassembly. *Automation Science and Engineering, IEEE Transactions on* 10 (07 2013), 496–502. doi:10.1109/TASE.2013.2245122
- DextaRobotics. 2026. . <https://www.dextarobotics.com/en-us>
- Haptx. 2026. . <https://haptx.com>
- He, Shiyu, Jing, Yunqi, Lu, Yiren, and Liu, Zhaoyu. 2023. Wearable Haptic Interfaces and Systems. *SHS Web Conf.* 157 (2023), 02024. doi:10.1051/shsconf/202315702024
- Fernando Hernandez-Gobertti, Ivan D. Kudyk, Raul Lozano, Giang T. Nguyen, and David Gomez-Barquero. 2025. Immersive Teleoperation via Collaborative Device-Agnostic Interfaces for Smart Haptics: A Study on Operational Efficiency and Cognitive Overflow for Industrial Assistive Applications. *Sensors* 25, 13 (2025). doi:10.3390/s25133993
- G. Jacucci, A. Bellucci, I. Ahmed, V. Harjunen, M. Spape, and N. Ravaja. 2024. Haptics in social interaction with agents and avatars in virtual reality: a systematic review. *Virtual Reality* 28,4 (2024). doi:10.1007/s10055-024-01060-6
- H. Jin, Y. Kim, W. Youm, Y. Min, S. Seo, C. Lim, C.-H. Hong, S. Kwon, Park G., S. Park, and H.J. Kim. 2022. Highly pixelated, untethered tactile interfaces for an ultra-flexible on-skin telehaptic system. *Flexible Electronics*, 6, 1 (2022).

- D. Lin, W. and Zhang, W.W. Lee, X. Li, Y. Hong, Q. Pan, R. Zhang, G. Peng, H.Z. Tan, Z. Zhang, L. Wei, and Z. Yang. 2022. Super-resolution wearable electrotactile rendering system. *Science Advances* 8,36 (2022).
- X. Liu, G. and Geng, L. Liu, and Y. Wang. 2019. Haptic based teleoperation with master-slave motion mapping and haptic rendering for space exploration. *Chinese Journal of Aeronautics* 32, 3 (2019), 723–736.
- Manus. 2026. . <https://www.manus-meta.com>
- A.K. Nassar, F. Al-Manaseer, L.M. Knowlton, and F. Tuma. 2021. Virtual reality (VR) as a simulation modality for technical skills acquisition. *Annals of Medicine and Surgery* 71 (2021).
- Marion Pontreau, Sylvain Bouchigny, Sabrina Panéels, and Sinan Haliyo. 2025a. WAL: Wearable Based on Active Lubrication for Virtual Reality. In *Haptics: Understanding Touch; Technology and Systems; Applications and Interaction*, Hiroyuki Kajimoto, Pedro Lopes, Claudio Pacchierotti, Cagatay Basdogan, Monica Gori, Betty Lemaire-Semail, and Maud Marchal (Eds.). Springer Nature Switzerland, Cham, 291–304. doi:10.1007/978-3-031-70061-3_24
- Marion Pontreau, Sabrina Panéels, Sylvain Bouchigny, and Sinan Haliyo. 2025b. Design and Evaluation of an Active Lubrication Brake for a Surgical Drilling Simulator. In *2025 IEEE World Haptics Conference (WHC)*. 1–14. doi:10.1109/WHC64065.2025.11123264
- SenseGlove. 2026. . <https://www.senseglove.com>
- Vivian Shen, Tucker Rae-Grant, Joe Mullenbach, Chris Harrison, and Craig Shultz. 2023. Fluid Reality: High-Resolution, Untethered Haptic Gloves using Electroosmotic Pump Arrays. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) (UIST '23). Association for Computing Machinery, New York, NY, USA, Article 8, 20 pages. doi:10.1145/3586183.3606771
- Shan-Yuan Teng, Yudai Tanaka, Alex Mazursky, and Pedro Lopes. 2026. Next Generation Wearable Haptics Should Balance Virtual & Real-world Fidelity. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*. Association for Computing Machinery, New York, NY, USA, Article 569, 19 pages. doi:10.1145/3772318.3790996
- Anastasios Theodoropoulos and Angeliki Antoniou. 2022. VR Games in Cultural Heritage: A Systematic Review of the Emerging Fields of Virtual Reality and Culture Games. *Applied Sciences* 12 (08 2022), 8476. doi:10.3390/app12178476
- Weart. 2026. . <https://weart.it>
- Biao Xie, Huimin Liu, Rawan Alghofaili, Yongqi Zhang, Yeling Jiang, Flavio Destri Lobo, Changyang Li, Wanwan Li, Haikun Huang, Mesut Akdere, Christos Mousas, and Lap-Fai Yu. 2021. A Review on Virtual Reality Skill Training Applications. *Frontiers in Virtual Reality* Volume 2 - 2021 (2021). doi:10.3389/frvir.2021.645153